

Physics Practice Questions — 150 Problems

Beginner (B1–B50): Level 1–2, 1–3 cards. Intermediate (I1–I50): Level 2–3, 2–5 cards. Advanced (A1–A50): Level 3–4, 4–10 cards.

Every [Cards: ...] tag is an authoritative reference to [honors-physics-expression-to-equation-vocabulary.md](#). Solutions in [physics-answers.md](#). Method: [physics-without-intuition.md](#).

BEGINNER — 50 Problems (Level 1–2, 1–3 cards)

B1–B10: Kinematics

- B1.** A car accelerates uniformly from rest at 3.0 m/s^2 for 8.0 s . How far does it travel? [Cards: 1, 4]
- B2.** A stone is dropped from a 45 m cliff. How long until it hits the ground? ($g = 10 \text{ m/s}^2$) [Cards: 1, 6]
- B3.** A ball is thrown straight up at 20 m/s . How high does it go? ($g = 10 \text{ m/s}^2$) [Cards: 13, 16]
- B4.** A cyclist at 12 m/s brakes to a stop with constant deceleration over 36 m . Find the deceleration. [Cards: 2, 4, 9]
- B5.** A car travels at constant velocity 25 m/s for 10 s . How far does it travel? [Card: 3]
- B6.** A ball is thrown horizontally at 15 m/s from a 20 m high cliff. How far from the base does it land? ($g = 10 \text{ m/s}^2$) [Cards: 1, 6, 12]
- B7.** A projectile launched at 40 m/s at 30° lands at the same height. Find time of flight. ($g = 10 \text{ m/s}^2$) [Cards: 12, 15]
- B8.** A rocket accelerates at 5.0 m/s^2 for 6.0 s from rest. Find final speed. [Cards: 1, 4, 9]
- B9.** A ball thrown vertically upward reaches max height 11.25 m . Initial speed? ($g = 10 \text{ m/s}^2$) [Cards: 13, 16, 62]
- B10.** A train at 30 m/s decelerates at 2.0 m/s^2 . Time to stop? [Cards: 2, 4, 7]

B11–B18: Forces & Newton's Laws

- B11.** A 10 kg box on a frictionless surface is pushed with 40 N . Find acceleration. [Cards: 23, 31]
- B12.** A 50 kg person stands on a scale in an elevator accelerating upward at 2.0 m/s^2 . Scale reading? ($g = 10 \text{ m/s}^2$) [Cards: 27, 44]
- B13.** A 2.0 kg block slides down a frictionless 30° incline. Find acceleration. ($g = 10 \text{ m/s}^2$) [Cards: 38, 39]
- B14.** A 5.0 kg block on a rough horizontal surface ($\mu_k = 0.40$) is pulled by 30 N . Find acceleration. ($g = 10 \text{ m/s}^2$) [Cards: 23, 27, 29]

B15. A 0.50 kg mass hangs from a spring ($k = 200 \text{ N/m}$). How far does the spring stretch? ($g = 10 \text{ m/s}^2$) [Cards: 25, 26, 42]

B16. Two masses (3.0 kg, 5.0 kg) hang over a frictionless pulley. Find acceleration. ($g = 10 \text{ m/s}^2$) [Cards: 33, 35, 36]

B17. A 4.0 kg block on a 37° incline ($\mu_k = 0.25$) slides from rest. Find acceleration. ($g = 10 \text{ m/s}^2$) [Cards: 1, 38, 40]

B18. A 1500 kg car rounds a flat curve ($R = 50 \text{ m}$, $\mu_s = 0.80$). Maximum speed without skidding? ($g = 10 \text{ m/s}^2$) [Cards: 30, 89]

B19–B25: Energy & Work

B19. A 2.0 kg block is lifted 3.0 m vertically. Work done by gravity? ($g = 10 \text{ m/s}^2$) [Cards: 26, 52]

B20. A 0.50 kg ball falls from rest from 20 m. Speed at impact? ($g = 10 \text{ m/s}^2$) [Cards: 1, 6, 61]

B21. A spring ($k = 500 \text{ N/m}$) is compressed 0.20 m. Energy stored? [Card: 58]

B22. A 1200 kg car at 15 m/s brakes to a stop. Work done by friction? [Cards: 2, 51, 56]

B23. A 0.10 kg rock is thrown upward at 30 m/s. Max height? ($g = 10 \text{ m/s}^2$) [Cards: 13, 60, 62]

B24. A motor lifts a 200 kg load 10 m in 5.0 s at constant speed. Power output? ($g = 10 \text{ m/s}^2$) [Cards: 3, 26, 67]

B25. A block slides 4.0 m on a rough surface ($\mu_k = 0.50$) before stopping. Initial speed? ($g = 10 \text{ m/s}^2$) [Cards: 2, 29, 51, 56]

B26–B32: Momentum & Collisions

B26. A 0.50 kg ball at 10 m/s strikes a wall and rebounds at 6.0 m/s. Impulse? [Card: 70]

B27. A 1200 kg car at 20 m/s rear-ends a 900 kg car at 15 m/s. They stick together. Find final speed. [Cards: 72, 73]

B28. A 0.020 kg bullet at 400 m/s embeds in a 2.0 kg block at rest on frictionless surface. Block's speed after? [Cards: 31, 72, 73]

B29. A 60 kg ice skater pushes a 40 kg skater. The 40 kg skater moves at 3.0 m/s. Speed of the 60 kg skater? [Cards: 77, 78]

B30. Two identical 0.20 kg pucks collide elastically, one initially at rest at 4.0 m/s. Final speed of the target? [Cards: 74, 75]

B31. A 3.0 kg object at 8.0 m/s explodes into two pieces: 1.0 kg at 12 m/s forward. Speed of 2.0 kg piece? [Card: 78]

B32. A 0.010 kg bullet at 300 m/s embeds in a 0.99 kg pendulum bob ($L = 1.0 \text{ m}$). Max height of swing? ($g = 10 \text{ m/s}^2$) [Cards: 72, 73, 79, 322]

B33–B38: Circular Motion & Gravitation

- B33.** A 0.50 kg ball on a 1.0 m string moves in a horizontal circle at 4.0 m/s. Centripetal force? [Cards: 83, 84]
- B34.** A satellite orbits Earth at $r = 7.0 \times 10^6$ m. Orbital speed? ($G = 6.67 \times 10^{-11}$, $M_E = 5.98 \times 10^{24}$) [Cards: 96, 97]
- B35.** A Ferris wheel ($R = 8.0$ m) rotates at 0.50 rad/s. Centripetal acceleration of a rider? [Card: 83]
- B36.** A car on a frictionless banked curve ($\theta = 25^\circ$, $R = 60$ m). Safe speed? ($g = 10$ m/s²) [Card: 88]
- B37.** Earth orbits the Sun at $r = 1.50 \times 10^{11}$ m. Period? ($G = 6.67 \times 10^{-11}$, $M_S = 1.99 \times 10^{30}$) [Cards: 96, 98]
- B38.** A ball on a 0.80 m string swings in a vertical circle. Minimum speed at top? ($g = 10$ m/s²) [Cards: 91, 92]

B39–B42: Rotational, SHM, Waves

- B39.** A solid disk ($I = \frac{1}{2}MR^2$, $M = 4.0$ kg, $R = 0.30$ m) experiences net torque 6.0 N m. Angular acceleration? [Cards: 103, 107]
- B40.** A 0.20 kg mass on a spring oscillates with period 0.50 s. Find k . [Card: 122]
- B41.** A simple pendulum ($L = 1.0$ m) has period? ($g = 10$ m/s²) [Card: 123]
- B42.** A guitar string ($L = 0.60$ m) has wave speed 300 m/s. Fundamental frequency? [Cards: 132, 142]

B43–B50: E&M, Thermo, Fluids, Optics, Modern

- B43.** Two point charges $+4.0 \mu\text{C}$ and $+6.0 \mu\text{C}$ are 0.30 m apart. Force magnitude? ($k = 9.0 \times 10^9$) [Card: 154]
- B44.** A 100 Ω resistor carries 0.50 A. Voltage across it? [Card: 171]
- B45.** Three 6.0 Ω resistors in parallel. Equivalent resistance? [Card: 173]
- B46.** An ideal gas at 2.0 atm, 300 K in a sealed container is heated to 450 K. New pressure? [Card: 220]
- B47.** Water ($\rho = 1000$ kg/m³) at depth 5.0 m. Gauge pressure? ($g = 10$ m/s²) [Card: 249]
- B48.** Light enters water ($n = 1.33$) from air at 40° to the normal. Refraction angle? [Card: 262]
- B49.** A photon has wavelength 500 nm. Energy in eV? ($hc = 1240$ eV nm) [Card: 284]
- B50.** A radioactive sample has half-life 8.0 days. What fraction remains after 24 days? [Card: 305]

INTERMEDIATE — 50 Problems (Level 2–3, 2–5 cards)

I1–I12: Mechanics Chains

- 11.** A 1500 kg car at 20 m/s brakes on a flat road ($\mu_k = 0.80$). Stopping distance? ($g = 10 \text{ m/s}^2$) [Cards: 2, 4, 27, 29]
- 12.** A 0.50 kg block slides from rest 2.0 m down a frictionless 37° incline, then 1.0 m on rough surface ($\mu_k = 0.40$). Total distance to stop on rough surface? ($g = 10 \text{ m/s}^2$) [Cards: 1, 4, 29, 38, 39]
- 13.** A pendulum bob (0.20 kg, $L = 1.0 \text{ m}$) released from rest at 60° . Speed at bottom? ($g = 10 \text{ m/s}^2$) [Cards: 1, 60, 61, 67]
- 14.** A 3.0 kg block compresses a spring ($k = 1200 \text{ N/m}$) 0.15 m on frictionless surface and is released. Speed when it leaves the spring? [Cards: 31, 58, 60]
- 15.** Two masses (4.0 kg, 6.0 kg) hang from a frictionless pulley, released from rest. Find acceleration. ($g = 10 \text{ m/s}^2$) [Cards: 1, 33, 35, 36]
- 16.** A 5.0 kg block on a 30° incline ($\mu_k = 0.30$) is connected by rope over a frictionless pulley to a hanging 3.0 kg mass. The 3.0 kg mass rises. Find acceleration. ($g = 10 \text{ m/s}^2$) [Cards: 23, 29, 33, 35, 38]
- 17.** A ball on a 0.80 m string swings in a vertical circle. Minimum speed at top? ($g = 10 \text{ m/s}^2$) [Cards: 91, 92]
- 18.** A 1200 kg car rounds an unbanked curve ($R = 40 \text{ m}$, $\mu_s = 0.70$). Maximum speed? ($g = 10 \text{ m/s}^2$) [Cards: 30, 89]
- 19.** A 0.10 kg mass on a spring has amplitude 0.050 m and $v_{\max} = 2.0 \text{ m/s}$. Find k . [Cards: 122, 126]
- 110.** A solid cylinder ($I = \frac{1}{2}mR^2$) rolls without slipping from rest down a 25° incline, traveling 3.0 m along the incline. Find speed at bottom. ($g = 10 \text{ m/s}^2$) [Cards: 1, 38, 60, 107, 113, 333]
- 111.** A 0.040 kg bullet at 350 m/s embeds in a 2.0 kg block on a frictionless surface. The block then compresses a spring ($k = 1500 \text{ N/m}$). Max compression? [Cards: 31, 42, 58, 72, 73, 321]
- 112.** A 2.0 kg block slides from rest 4.0 m down a frictionless 30° incline, then collides elastically with a stationary 3.0 kg block on a frictionless horizontal surface. Final velocities? ($g = 10 \text{ m/s}^2$) [Cards: 1, 38, 39, 60, 72, 74]

I13–I20: E&M & Circuits

- 113.** An electron accelerated from rest through 2000 V enters $B = 0.15 \text{ T}$ perpendicular. Find radius of circular path. ($m_e = 9.11 \times 10^{-31} \text{ kg}$, $e = 1.60 \times 10^{-19} \text{ C}$) [Cards: 1, 193, 195, 338]
- 114.** Resistors $R_1 = 6.0 \Omega$ and $R_2 = 3.0 \Omega$ in parallel, with $R_3 = 4.0 \Omega$ in series. 12 V battery. Find current through R_2 . [Cards: 171, 172, 173]
- 115.** A parallel-plate capacitor has $A = 0.050 \text{ m}^2$, $d = 1.0 \text{ mm}$. Capacitance? ($\epsilon_0 = 8.85 \times 10^{-12}$) [Card: 183]
- 116.** A velocity selector has $E = 3.0 \times 10^4 \text{ N/C}$ and $B = 0.060 \text{ T}$. Speed of undeflected ions? [Card: 196]
- 117.** A long straight wire carries 5.0 A. Magnetic field at 0.10 m from the wire? ($\mu_0 = 4\pi \times 10^{-7}$) [Card: 202]

- I18.** A 100-turn coil (area 0.020 m^2) rotates at 60 rev/s in $B = 0.40 \text{ T}$. Maximum induced emf? [Card: 210]
- I19.** Two capacitors $C_1 = 4.0 \text{ }\mu\text{F}$ and $C_2 = 6.0 \text{ }\mu\text{F}$ in series. Equivalent capacitance? [Card: 186]
- I20.** An RL circuit: $L = 0.50 \text{ H}$, $R = 10 \text{ }\Omega$ connected to 20 V . Time constant? Current after 0.10 s (from zero)? [Cards: 171, 215]

I21–I27: Thermo & Fluids

- I21.** $n = 0.50 \text{ mol}$ of ideal gas expands isothermally at 400 K from 2.0 L to 6.0 L . Work done? ($R = 8.31$) [Cards: 217, 221, 234]
- I22.** A Carnot engine operates between 500 K and 300 K . Absorbs 1000 J from hot reservoir. Work output? Efficiency? [Cards: 239, 240]
- I23.** Monatomic gas ($n = 1.0$ at 300 K) compressed adiabatically to half volume. Final temperature? ($\gamma = 5/3$) [Cards: 235, 236]
- I24.** 0.20 kg ice at 0°C is dropped into 0.50 kg water at 40°C . Final temperature? ($c_w = 4186$, $L_f = 3.34 \times 10^5$) [Cards: 226, 228, 231]
- I25.** Water flows through a pipe narrowing from $D_1 = 6.0 \text{ cm}$ to $D_2 = 3.0 \text{ cm}$. $v_1 = 2.0 \text{ m/s}$. Find v_2 and ΔP (horizontal). ($\rho = 1000 \text{ kg/m}^3$) [Cards: 254, 255]
- I26.** A 500 cm^3 object ($\rho = 800 \text{ kg/m}^3$) floats in water. Volume submerged? [Cards: 247, 253]
- I27.** Water exits a hole 0.50 m below the surface of a tank, 1.5 m above ground. Horizontal range? ($g = 10 \text{ m/s}^2$) [Cards: 12, 14, 256, 355]

I28–I35: Waves, Optics, Modern

- I28.** An open pipe ($L = 1.2 \text{ m}$) sounds its 3rd harmonic. Frequency? ($v = 340 \text{ m/s}$) [Cards: 132, 140]
- I29.** A train at 40 m/s blows 600 Hz horn. Frequency heard by a stationary observer as the train approaches? ($v = 340 \text{ m/s}$) [Card: 145]
- I30.** Two tuning forks at 440 Hz and 444 Hz sound together. Beat frequency? [Card: 147]
- I31.** A double-slit: $d = 0.30 \text{ mm}$, $L = 1.5 \text{ m}$, $\lambda = 600 \text{ nm}$. Fringe spacing? [Card: 276]
- I32.** A converging lens ($f = 12 \text{ cm}$). Object at 18 cm . Image distance? Is it real? [Cards: 266, 272]
- I33.** Light in water ($n = 1.33$) strikes the water-air interface. Critical angle? [Card: 264]
- I34.** Photoelectric effect: metal with $\phi = 2.3 \text{ eV}$ illuminated by $\lambda = 400 \text{ nm}$. K_{max} ? ($hc = 1240 \text{ eV nm}$) [Cards: 284, 285]
- I35.** Electron ($v = 2.0 \times 10^6 \text{ m/s}$). de Broglie wavelength? ($h = 6.63 \times 10^{-34}$, $m_e = 9.11 \times 10^{-31}$) [Card: 290]

I36–I42: Rotational & SHM

- I36.** A solid sphere ($I = \frac{2}{5}mR^2$) rolls without slipping from rest down a **3.0 m** high incline. Speed at bottom? ($g = 10 \text{ m/s}^2$) [Cards: 1, 60, 108, 113, 116, 333]
- I37.** A massive pulley ($M = 2.0 \text{ kg}$, solid cylinder, $R = 0.10 \text{ m}$) connects a **5.0 kg** block on a frictionless horizontal surface to a hanging **3.0 kg** mass. Find acceleration. ($g = 10 \text{ m/s}^2$) [Cards: 23, 31, 33, 35, 107, 334]
- I38.** A **0.30 kg** mass on a spring ($k = 200 \text{ N/m}$) oscillates with amplitude **0.080 m**. Total energy? v_{max} ? [Cards: 122, 126, 129]
- I39.** A physical pendulum: uniform rod ($L = 0.50 \text{ m}$, pivot at end). Period? ($g = 10 \text{ m/s}^2$, $I_{\text{end}} = \frac{1}{3}ML^2$) [Cards: 111, 124]
- I40.** A spinning ice skater ($I = 4.0 \text{ kg m}^2$, $\omega = 3.0 \text{ rad/s}$) pulls arms in ($I = 2.0 \text{ kg m}^2$). New ω ? [Card: 118]
- I41.** Torque $\tau = 20 \text{ N m}$ acts on a disk ($I = 5.0 \text{ kg m}^2$) for **3.0 s** from rest. Final ω ? [Cards: 1, 103, 119]
- I42.** A yo-yo ($m = 0.15 \text{ kg}$, $R = 0.040 \text{ m}$, $I = \frac{1}{2}mR^2$) falls **1.0 m** from rest while unwinding. Speed at bottom? ($g = 10 \text{ m/s}^2$) [Cards: 1, 60, 107, 113, 116, 335]

I43–I50: Mixed Domains

- I43.** **0.50 kg** block slides **3.0 m** down a **37°** incline ($\mu_k = 0.20$). At the bottom, it compresses a spring ($k = 1500 \text{ N/m}$). Max compression? ($g = 10 \text{ m/s}^2$) [Cards: 1, 4, 38, 40, 58, 60]
- I44.** A **0.020 kg** bullet at **300 m/s** embeds in a **1.5 kg** block. The block slides **2.0 m** on a rough surface ($\mu_k = 0.25$) before stopping. Verify consistency. ($g = 10 \text{ m/s}^2$) [Cards: 2, 29, 51, 56, 72, 73]
- I45.** A **1000 kg** satellite orbits at **300 km** altitude ($r = 6.67 \times 10^6 \text{ m}$). Orbital period? ($G = 6.67 \times 10^{-11}$, $M_E = 5.98 \times 10^{24}$) [Cards: 96, 98]
- I46.** $R_1 = 10 \Omega$, $R_2 = 20 \Omega$ in parallel, connected to **12 V** battery. Total power dissipated? [Cards: 171, 173, 178]
- I47.** A proton accelerates from rest through **5000 V**, enters $B = 0.80 \text{ T}$ perpendicular. Radius? ($m_p = 1.67 \times 10^{-27} \text{ kg}$, $e = 1.60 \times 10^{-19} \text{ C}$) [Cards: 1, 193, 195, 338]
- I48.** A sound wave ($f = 1000 \text{ Hz}$) in air at **20°C** ($v = 343 \text{ m/s}$). Wavelength? The same wave in helium ($M = 0.004 \text{ kg/mol}$, $\gamma = 1.67$, $T = 293 \text{ K}$, $R = 8.31$)? [Cards: 132, 150, 354]
- I49.** Hydrogen electron drops from $n = 6$ to $n = 2$. Wavelength? Series? ($R = 1.097 \times 10^7 \text{ m}^{-1}$) [Cards: 292, 294]
- I50.** A **2.0 kg** block on a horizontal spring ($k = 300 \text{ N/m}$) oscillates with amplitude **0.10 m**. At $x = 0$, a **1.0 kg** putty drops on and sticks. New amplitude? [Cards: 72, 73, 122, 126, 129]

ADVANCED — 50 Problems (Level 3–4, 4–10 cards)

A1–A15: Mechanics Multi-Domain Synthesis

- A1.** A 0.030 kg bullet at 500 m/s embeds in a 1.5 kg block on a frictionless surface. The block compresses a spring ($k = 2000\text{ N/m}$). Max compression? [Cards: 1, 31, 42, 58, 72, 73, 321]
- A2.** A 2.0 kg block slides from rest 4.0 m down a frictionless 30° incline, then collides elastically with a stationary 3.0 kg block. Find velocities after collision. ($g = 10\text{ m/s}^2$) [Cards: 1, 38, 39, 60, 72, 74, 75]
- A3.** A 0.020 kg bullet at 400 m/s embeds in a 2.0 kg pendulum bob ($L = 1.2\text{ m}$). Find max swing angle. ($g = 10\text{ m/s}^2$) [Cards: 60, 67, 72, 73, 79, 322, 327]
- A4.** Roller coaster from rest at height h enters loop ($R = 8.0\text{ m}$). Minimum h ? ($g = 10\text{ m/s}^2$) [Cards: 1, 60, 65, 66, 91, 92, 328]
- A5.** Solid sphere ($I = \frac{2}{5}mR^2$) rolls without slipping from rest down an incline of height 3.0 m . Speed at bottom? Compare to sliding. ($g = 10\text{ m/s}^2$) [Cards: 1, 38, 60, 108, 113, 116, 333]
- A6.** A 5.0 kg block on a 37° incline ($\mu_k = 0.25$) is connected by rope over a massive pulley ($M = 2.0\text{ kg}$, $I = \frac{1}{2}MR^2$, $R = 0.10\text{ m}$) to a hanging 8.0 kg mass. Block slides UP. Find acceleration. ($g = 10\text{ m/s}^2$) [Cards: 23, 29, 33, 35, 38, 107, 334]
- A7.** Bowling ball ($I = \frac{2}{5}mR^2$) initially sliding at 8.0 m/s on a lane ($\mu_k = 0.12$). Distance to pure rolling? ($g = 10\text{ m/s}^2$) [Cards: 29, 103, 107, 113, 337]
- A8.** Spring ($k = 800\text{ N/m}$) compressed 0.10 m launches a 0.20 kg block. Block slides 1.5 m on rough surface ($\mu_k = 0.30$) then flies off a 2.0 m high table. Landing distance? ($g = 10\text{ m/s}^2$) [Cards: 1, 6, 12, 29, 31, 58, 60]
- A9.** Two blocks (3.0 kg , 2.0 kg) connected by a spring ($k = 600\text{ N/m}$) compressed 0.15 m on a frictionless surface, released from rest. Find speeds after separation. [Cards: 1, 31, 42, 58, 60, 72, 78, 325]
- A10.** Pendulum ($m = 0.50\text{ kg}$, $L = 1.0\text{ m}$) released from horizontal. At bottom, string hits a peg 0.60 m below pivot. Find bottom speed and tension just after hitting peg. ($g = 10\text{ m/s}^2$) [Cards: 1, 60, 61, 67, 83, 84, 91]
- A11.** A 1.0 kg block slides from rest 5.0 m down a frictionless 37° incline, then 2.0 m on rough surface ($\mu_k = 0.20$), then compresses a spring ($k = 1500\text{ N/m}$). Max compression? ($g = 10\text{ m/s}^2$) [Cards: 1, 4, 29, 38, 39, 56, 58, 60, 63]
- A12.** A 0.050 kg rubber ball dropped from 4.0 m rebounds to 3.0 m . Coefficient of restitution? ($g = 10\text{ m/s}^2$) [Cards: 1, 6, 61, 76]
- A13.** A 4.0 kg block on a rough 30° incline ($\mu_k = 0.20$) slides 3.0 m from rest, then collides inelastically ($e = 0$) with a 2.0 kg block at the bottom. Speed after collision? ($g = 10\text{ m/s}^2$) [Cards: 1, 4, 29, 38, 40, 72, 73]
- A14.** A 1200 kg car at 25 m/s rear-ends a 900 kg car at 15 m/s . They stick together. Speed after? Energy lost? [Cards: 56, 72, 73]
- A15.** A solid cylinder rolls without slipping up a 20° incline with initial speed 6.0 m/s at the bottom. How far up the incline? ($g = 10\text{ m/s}^2$) [Cards: 2, 38, 57, 60, 107, 113, 114, 116]

A16–A25: E&M Cross-Domain

- A16.** Electron accelerated from rest through 2000 V enters $B = 0.15$ T perpendicular. Radius? ($m_e = 9.11 \times 10^{-31}$ kg, $e = 1.60 \times 10^{-19}$ C) [Cards: 1, 193, 195, 338]
- A17.** Velocity selector: $E = 3.0 \times 10^4$ N/C, B_1 unknown. Singly charged ions pass undeflected, then $B_2 = 0.50$ T gives $r = 0.12$ m. $m = 2.0 \times 10^{-26}$ kg. Find B_1 . [Cards: 193, 195, 196, 339]
- A18.** Bar ($m = 0.30$ kg, $L = 0.25$ m) slides down frictionless rails at 40° in $B = 0.50$ T horizontal. $R = 2.0$ Ω at bottom. Terminal velocity? ($g = 10$ m/s²) [Cards: 25, 29, 38, 171, 194, 207, 209, 345]
- A19.** A circular loop ($R = 0.050$ m, $N = 20$) has its plane perpendicular to $B = 0.30$ T. The field drops to zero in 0.020 s. Average induced emf? [Cards: 206, 207, 208]
- A20.** An LC circuit: $L = 0.10$ H, $C = 2.0$ μ F. Initially C has 50 μ C. Maximum current? [Cards: 210, 214, 216]
- A21.** A copper bar ($L = 0.50$ m) moves at 10 m/s perpendicular to $B = 0.40$ T. Induced emf drives current through a 5.0 Ω resistor. Power dissipated? [Cards: 171, 178, 207, 209]
- A22.** A long solenoid ($n = 1000$ m⁻¹, $I = 3.0$ A) has a 100-turn coil (area 0.0050 m²) wrapped around its center. Current reverses in 0.010 s. Average induced emf? [Cards: 204, 206, 207]
- A23.** Two long parallel wires 0.10 m apart carry $I_1 = 5.0$ A, $I_2 = 3.0$ A in the same direction. Force per meter? Attract or repel? ($\mu_0 = 4\pi \times 10^{-7}$) [Card: 198]
- A24.** An electron ($K = 10$ keV) enters $B = 0.60$ T at 30° to the field. Find pitch of helical path. ($m_e = 9.11 \times 10^{-31}$ kg, $e = 1.60 \times 10^{-19}$ C) [Cards: 56, 193, 195]
- A25.** A proton ($v = 2.0 \times 10^7$ m/s) enters crossed $\vec{E} = 1.0 \times 10^5$ N/C (up) and $\vec{B} = 0.020$ T (into page). Net force (magnitude and direction)? ($e = 1.60 \times 10^{-19}$ C) [Cards: 159, 193]

A26–A33: Thermo, Fluids, Waves, Optics Cross-Domain

- A26.** $n = 0.50$ mol gas expands isothermally at 400 K from 2.0 L to 6.0 L. Work and heat? ($R = 8.31$) [Cards: 217, 221, 234]
- A27.** Monatomic gas ($n = 1.0$, $T_1 = 300$ K) compressed adiabatically to half volume. Final T and work done by gas? ($\gamma = 5/3$, $C_V = \frac{3}{2}R$, $R = 8.31$) [Cards: 235, 236]
- A28.** Water exits a hole 0.50 m below surface of a tank, 1.5 m above ground. Horizontal range? ($g = 10$ m/s²) [Cards: 12, 14, 256, 355]
- A29.** A Venturi meter: $D_1 = 8.0$ cm, $D_2 = 4.0$ cm, $\Delta P = 1500$ Pa (water, $\rho = 1000$ kg/m³). Flow speed v_1 ? Volume flow rate? [Cards: 254, 255, 257]
- A30.** A sound wave in helium ($M = 0.004$ kg/mol, $\gamma = 1.67$, $T = 293$ K). Speed? ($R = 8.31$) Compare to air ($v = 343$ m/s). [Cards: 150, 354]
- A31.** A double-slit ($d = 0.30$ mm, $L = 1.5$ m) illuminated with white light (400–700 nm). Which visible λ has its 2nd-order maximum coinciding with the 3rd-order minimum of another λ ? [Cards: 275, 276]
- A32.** A thin film ($n = 1.40$, $t = 250$ nm) on glass ($n = 1.50$) in air. Which visible wavelengths are strongly reflected? [Cards: 132, 279]

A33. A converging lens ($f = 12 \text{ cm}$) and diverging lens ($f = -8.0 \text{ cm}$) are 4.0 cm apart. Object 18 cm before converging lens. Final image position? [Cards: 266, 272, 273]

A34–A42: Multi-Domain Synthesis (3+ domains)

A34. Spring ($k = 1200 \text{ N/m}$) compressed 0.20 m launches 0.50 kg block on frictionless surface. Block slides up 30° incline ($d = 4.0 \text{ m}$ along incline). At top, block flies off as projectile from 5.0 m height. Landing distance? ($g = 10 \text{ m/s}^2$) [Cards: 1, 6, 12, 31, 38, 57, 58, 60]

A35. Gas in cylinder ($n = 0.30 \text{ mol}$, $T = 500 \text{ K}$) expands isothermally, pushing a piston ($A = 0.015 \text{ m}^2$) through 0.25 m . Piston is connected to a spring ($k = 1500 \text{ N/m}$) initially unstretched. Find final spring compression and work done by gas. ($R = 8.31$) [Cards: 42, 217, 221, 234, 350, 351]

A36. A neutron ($m = 1.67 \times 10^{-27} \text{ kg}$) with de Broglie $\lambda = 0.10 \text{ nm}$ enters $B = 0.50 \text{ T}$ perpendicular. Orbital radius? Does a neutron orbit in a B-field? ($h = 6.63 \times 10^{-34}$) [Cards: 193, 195, 290]

A37. A 2.0 kg block slides from rest 6.0 m down a 37° incline ($\mu_k = 0.20$). At bottom, inelastic collision with 3.0 kg block at rest. Both then compress spring ($k = 2500 \text{ N/m}$). Max compression? ($g = 10 \text{ m/s}^2$) [Cards: 1, 4, 29, 38, 40, 56, 58, 60, 72, 73, 321]

A38. A 0.50 kg pendulum ($L = 0.80 \text{ m}$) pulled to horizontal, released. At bottom, bob enters a magnetic field region ($B = 2.0 \text{ T}$, perpendicular to motion). The bob is a conducting sphere ($R = 0.020 \text{ m}$). Eddy currents provide damping. Qualitative: what happens? ($g = 10 \text{ m/s}^2$) [Cards: 1, 60, 67, 83, 193, 207, 346]

A39. Capacitor ($100 \mu\text{F}$) charged to 200 V discharges through a 10Ω resistor. The resistor is immersed in 0.10 kg water ($c = 4186 \text{ J/(kg K)}$). Temperature rise? (Assume all energy becomes heat.) [Cards: 56, 171, 185, 188, 226, 372]

A40. A 1000 kg satellite at 300 km altitude ($r = 6.67 \times 10^6 \text{ m}$) fires thrusters doing $1.0 \times 10^8 \text{ J}$ work to raise orbit. New altitude? ($G = 6.67 \times 10^{-11}$, $M_E = 5.98 \times 10^{24}$) [Cards: 96, 97, 98, 101, 332]

A41. A proton and electron, both at rest, are released 1.0 cm apart. Find speeds when separation is 2.0 cm . ($k = 8.99 \times 10^9$, $e = 1.60 \times 10^{-19} \text{ C}$, $m_p = 1.67 \times 10^{-27} \text{ kg}$, $m_e = 9.11 \times 10^{-31} \text{ kg}$) [Cards: 1, 56, 60, 69, 72, 154, 161]

A42. A spaceship travels to Alpha Centauri (4.3 light-years) at $0.95c$. Trip duration according to Earth clocks? Ship clocks? ($c = 3.00 \times 10^8 \text{ m/s}$) [Cards: 296, 297, 298, 299]

A43–A50: Olympiad-Level (5+ cards, bridge chains)

A43. Spring ($k = 3000 \text{ N/m}$) compressed 0.15 m launches 0.40 kg block on rough surface ($\mu_k = 0.25$) for 1.0 m . Block then slides frictionlessly, compressing a second spring ($k = 2000 \text{ N/m}$). Max compression of second spring? ($g = 10 \text{ m/s}^2$) [Cards: 1, 29, 31, 42, 51, 58, 60, 63, 66]

A44. A solar sail ($A = 1000 \text{ m}^2$, $m = 10 \text{ kg}$) at Earth's orbit receives $I = 1400 \text{ W/m}^2$ (perfect reflection). Acceleration? Speed after 1 day? ($c = 3.0 \times 10^8 \text{ m/s}$) [Cards: 1, 4, 23, 56, 284, 288, 365, 371]

A45. An α -particle ($m = 6.64 \times 10^{-27} \text{ kg}$, $q = 2e$) and a proton ($m_p = 1.67 \times 10^{-27} \text{ kg}$, $q = e$) are both accelerated through 1000 V and enter the same $B = 0.50 \text{ T}$ perpendicular. Ratio of orbital radii r_α/r_p ? [Cards: 1, 193, 195, 338, 340]

- A46.** A 0.10 kg aluminum block ($c = 900 \text{ J}/(\text{kg K})$) at 200°C is placed on a 2.0 kg ice block at 0°C . How much ice melts? ($L_f = 3.34 \times 10^5 \text{ J/kg}$) [Cards: 226, 228, 231]
- A47.** A tuning fork ($f = 440 \text{ Hz}$) is struck, held over an open pipe of adjustable length. Minimum pipe length for resonance? ($v = 340 \text{ m/s}$) [Cards: 132, 140, 142]
- A48.** A 0.50 kg mass on a spring ($k = 200 \text{ N/m}$) in water experiences damping ($b = 5.0 \text{ kg/s}$). Damped frequency? Time for amplitude to halve? [Cards: 122, 130, 143]
- A49.** A radioactive source ($T_{1/2} = 10 \text{ min}$) produces 1000 counts/s at $t = 0$. After how long will the count rate be 125 counts/s? [Card: 305]
- A50.** A pogo stick: spring ($k = 5000 \text{ N/m}$) compressed 0.15 m under a 60 kg rider. Rider+stick ($m = 62 \text{ kg}$) launched vertically. Max height above uncompressed spring position? ($g = 10 \text{ m/s}^2$) [Cards: 1, 16, 42, 57, 58, 60, 62, 66, 375]